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- **Problem Statement ID –** 97
- **Problem Statement Title –** AI-Driven voice Assistant for livelihood Mapping and NSQF-Aligned Skilling Recommendations for SC Communities under GIA component of PM-AJAY
- **Organization –** Ministry of Social Justice and Empowerment (MoSJE)
- **Theme –** Agriculture, FoodTech & Rural Development
- **PS Category –** Software
- **Team ID –**
- **Team Name (Registered on portal) –** Jeevikasetu



AI-Powered Voice-to-Livelihood Platform

From Voice → Skills → Better Livelihood Opportunities

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2 Proposed Solution

From Voices to Skills to Better Livelihood Opportunities"

- We will develop a voice-first, AI-powered platform to help beneficiaries access suitable skilling and livelihood opportunities.
- The system will interact in local languages and create a digital livelihood profile from simple conversations.
- It will analyse existing skills, identify skill gaps and recommend NSQF-aligned training pathways.
- The platform will provide information on nearby training centers and local opportunities.
- Real-time tracking will show progress from training to certification and livelihood.
- A central dashboard will provide insights for authorities to plan and improve programs.

How It Works



Illustrative Beneficiary Journey



"One conversation can open a brighter future."

3. TECHNOLOGY APPROACH

System Architecture and Implementation Flow

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1. Beneficiary Input
- Voice / Text input
- Local language support
- Mobile App (Android / PWA)

API / WebSocket

2. Voice & Language Processing
- Speech-to-Text (STT)
- Language Detection
- Text-to-Speech (TTS)

3. AI / NLP Engine
- Intent Recognition
- Information Extraction
- Profile Generation
- Conversation Management

4. Livelihood Profile
- Skills
- Education
- Experience
- Aspirations
- Location
- Preferences

5. Skill Gap Analysis
- Match with required skills
- Identify missing skills

6. NSQF Knowledge Base
- Verified job roles & qualifications
- Skill requirements
- Training / RPL pathways
- Updated from official sources

7. Local Opportunity Data
- Nearby training centres
- Workshops / Programs
- Employment opportunities
- Local demand & industry data

8. Recommendation Engine
- AI + rule-based matching
- Personalized pathway
- Training / RPL recommendations
- Explainable suggestions

9. Output to Beneficiary (Mobile App)
- Recommended pathway
- Nearby training / workshops
- Relevant job opportunities
- Reason for recommendation

10. Organization / Admin Portal
- Post training / job opportunities
- Find matching beneficiaries
- Monitor progress
- District / block level analytics

Dashboard & Reports

11. Database
- PostgreSQL (with PostGIS)
- User profiles, skills, opportunities
- Progress and outcomes

12. Outcome Tracking
- Training enrollment
- Certification status
- Employment / Self-employment
- Impact analytics

Continuous Improvement (Feedback Loop)

Technologies Used

Mobile App	Flutter (Android / PWA)
Web Portal	React / Next.js
Backend	Python + FastAPI
AI / NLP	LLM + LangChain
Voice (STT/TTS)	Bhashini / Whisper (Indian Languages)
Database	PostgreSQL (with PostGIS)
Vector Database	pgvector / Chroma
Maps & Visualization	Leaflet.js / Mapbox
Analytics & Charts	Recharts / Chart.js
Authentication	JWT / Role-Based Access
Hosting	AWS / Azure / GCP
Version Control	GitHub

Key Highlights

- Voice-first and multilingual
- AI-powered personalized recommendations
- Integrates verified NSQF and local opportunity data
- Supports both beneficiaries and organizations
- Scalable, secure and modular architecture

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4. FEASIBILITY AND VIABILITY

Practical • Scalable • Sustainable

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Feasibility Analysis

The solution can be implemented using existing technologies, data sources and government ecosystem.

Technical Feasibility

- Built using widely available technologies (Flutter, React, Python, cloud, open APIs).
- Can integrate with government platforms (e.g., BHASHINI, NSQF, PM-AJAY).
- Can be developed as a working prototype within the hackathon timeline.

Operational Feasibility

- Simple and intuitive interface (voice-first, local languages).
- Minimal training required for beneficiaries and officials.
- Can be deployed through existing CSCs, training centres and district offices.

Economic Feasibility

- Cost-effective using open-source tools and government-enabled services.
- Reduces manual counselling effort and improves resource targeting.
- Scalable to multiple districts and states with minimal additional cost.

Legal & Policy Feasibility

- Compliant with data privacy and security norms.
- Aligns with government initiatives (PM-AJAY, NSQF, skilling and livelihood programs).
- Can be deployed in a phased manner with required approvals.

Challenges and Risks

Key challenges that may affect successful implementation.

Data Privacy & Security	Handling sensitive beneficiary data and ensuring secure storage and access.
Integration with Existing Systems	Alignment with government platforms and heterogeneous datasets.
User Adoption	Digital literacy gaps and initial resistance from beneficiaries and officials.
Data Quality	Incomplete or inconsistent data may affect accuracy of recommendations.

Strategies for Overcoming Challenges

Our approach to mitigate risks and ensure long-term success.

Strong encryption and secure data management	Use standard APIs and modular architecture	Conduct training and awareness programs	Continuous monitoring, feedback and iterative improvement

5. IMPACT AND BENEFITS

Empowered Individuals | Stronger Communities | Inclusive Growth

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For Beneficiaries

- Access to personalized career guidance
- Relevant training and livelihood opportunities
- Improved employability and income
- Voice-first, local language support

For Society

- Inclusive growth of SC communities
- Reduced unemployment and underemployment
- Social empowerment through skills
- Stronger and more resilient communities

For Organizations & Training Providers

- Easier identification of eligible candidates
- Better utilization of training capacity
- Targeted outreach to relevant beneficiaries
- Data insights for program design

IMPACT & BENEFITS

Skilled Communities
Stronger India

For Government

- Data-driven planning and policy making
- Identification of local skill gaps
- Better targeting of training programs
- Tracking outcomes and impact (PM-AJAY)

Economic Impact

- Higher employment and self-employment
- Increased income for beneficiaries
- Reduced dependence on manual/ informal work
- Contribution to local and national economy

Environmental Impact

- Promotion of green jobs and sustainable livelihoods
- Skill development in renewable and climate-resilient sectors
- Support for local, sustainable economic activities

"From Skills to Opportunities, From Communities to a Stronger India."

Sustainability & Revenue Model

Government Support
Implementation through government programs (e.g., PM-AJAY, NSQF, skilling initiatives).

Partnerships
Collaboration with training providers, industry partners and NGos.

Premium Analytics
Advanced dashboards and data insights for large organizations and research institutions.

Custom Integration
Integration with existing government and enterprise systems (API-based).

Grants & CSR Support
Funding from CSR initiatives and development programs.

Future Value-added Services
Additional services such as advanced skilling pathways, assessment tools and labor market insights.

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6. RESEARCH AND REFERENCES

Evidence Driven | Built on Existing Initiatives | Aligned with National Goals

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Innovation and Uniqueness

AI-powered personalized pathways
Uses AI to understand skills, identify gaps and recommend NSQF-aligned training and opportunities.

Voice-first and multilingual
Accessible to beneficiaries in local languages, including low-literacy users.

Two-sided platform
Connects beneficiaries with training providers, work opportunities and government programmes.

Data-driven decision making
Provides real-time insights for authorities on community skill gaps and training demand.

Leverages existing ecosystem
Integrates with national initiatives (PM-AJAY, NSQF, BHASHINI, etc.) for better reach and scalability.

Research Sources

- Government reports and documents related to skill development and livelihood (e.g., PM-AJAY, NSQF, Skill India).
- Case studies on skilling, RPL and employment models for marginalized communities.
- Technical documentation on AI, voice technologies (e.g., BHASHINI) and recommendation systems.
- Reports on local labour market trends and district-level skill demand.
- Articles and research papers on inclusive growth, digital skilling and social impact.
- Consultation with domain experts, NGOs and training providers.

References

- PM-AJAY Official Guidelines and Reports.
- NSQF Framework and Qualification Packs (NCVET).
- Skill India and PMKVY documentation.
- BHASHINI (Indian Language Technology) resources.
- Government labour market and employment reports (e.g., PLFS, Periodic Labour Force Survey).
- Research papers on AI for skilling and inclusive development.
- Relevant policy documents on SC welfare, livelihood and rural development.
- Websites and publications from MoSJE, MSDE, NCVET and other relevant organizations.